

In [3]: `pip install mysql-connector-python`

Requirement already satisfied: mysql-connector-python in c:\users\admin\anaconda3\lib\site-packages (8.2.0)Note: you may need to restart the kernel to use updated package s.

Requirement already satisfied: protobuf<=4.21.12,>=4.21.1 in c:\users\admin\anaconda3\lib\site-packages (from mysql-connector-python) (4.21.12)

In [53]: `# Importing module
import mysql.connector
import pandas as pd
import seaborn as sns
from matplotlib import pyplot as plt`

In [68]: `import mysql.connector
from mysql.connector import errorcode
try:
 cnx = mysql.connector.connect(host='127.0.0.1',
 database='superstore',
 user='root',
 password='Sri@199789')
except mysql.connector.Error as err:
 if err.errno == errorcode.ER_ACCESS_DENIED_ERROR:
 print("Something is wrong with your user name or password")
 elif err.errno == errorcode.ER_BAD_DB_ERROR:
 print("Database does not exist")
 else:
 print(err)
else:
 print("Sucessfully connected")
 mycursor = cnx.cursor()`

Sucessfully connected

In [69]: `mycursor.execute("SELECT Mode_Of_Payment, SUM(amount) AS Revenue, ROUND(100 * SUM(amou
result1 = pd.DataFrame(mycursor.fetchall(), columns=['Mode_Of_Payment', 'Revenue', 'Re`

In [70]: `print(result1)`

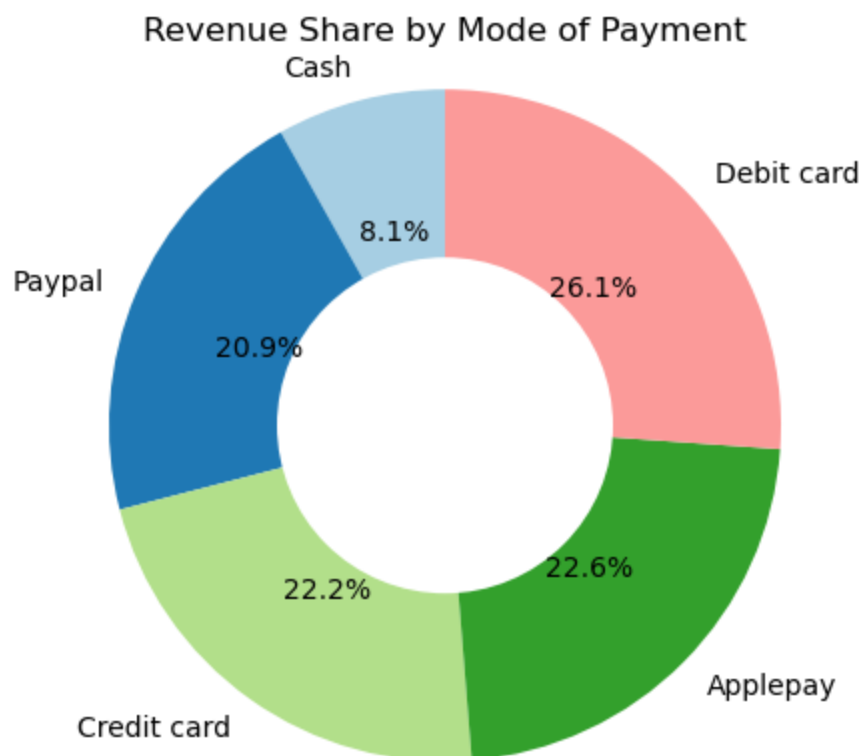
	Mode_Of_Payment	Revenue	Revenue_Share(%)
0	Cash	4636.0	8.13
1	Paypal	11926.0	20.91
2	Credit card	12688.0	22.24
3	Applepay	12891.0	22.60
4	Debit card	14906.0	26.13

In [71]: `# Visualize the data with a donut chart
labels = result1['Mode_Of_Payment']
sizes = result1['Revenue_Share(%)']
colors = plt.cm.Paired(range(len(labels)))

plt.pie(sizes, labels=labels, autopct='%1.1f%%', startangle=90, colors=colors)
centre_circle = plt.Circle((0, 0), 0.50, fc='white')
fig = plt.gcf()
fig.gca().add_artist(centre_circle)

plt.axis('equal') # Equal aspect ratio ensures that pie is drawn as a circle.`

```
plt.title('Revenue Share by Mode of Payment')
plt.show()
```



```
In [73]: mycursor.execute("SELECT C.Customer_ID,C.First_Name,C.Last_Name,SUM(P.Amount) AS Total_spend_$
# Fetch the results
result1_5 = mycursor.fetchall()

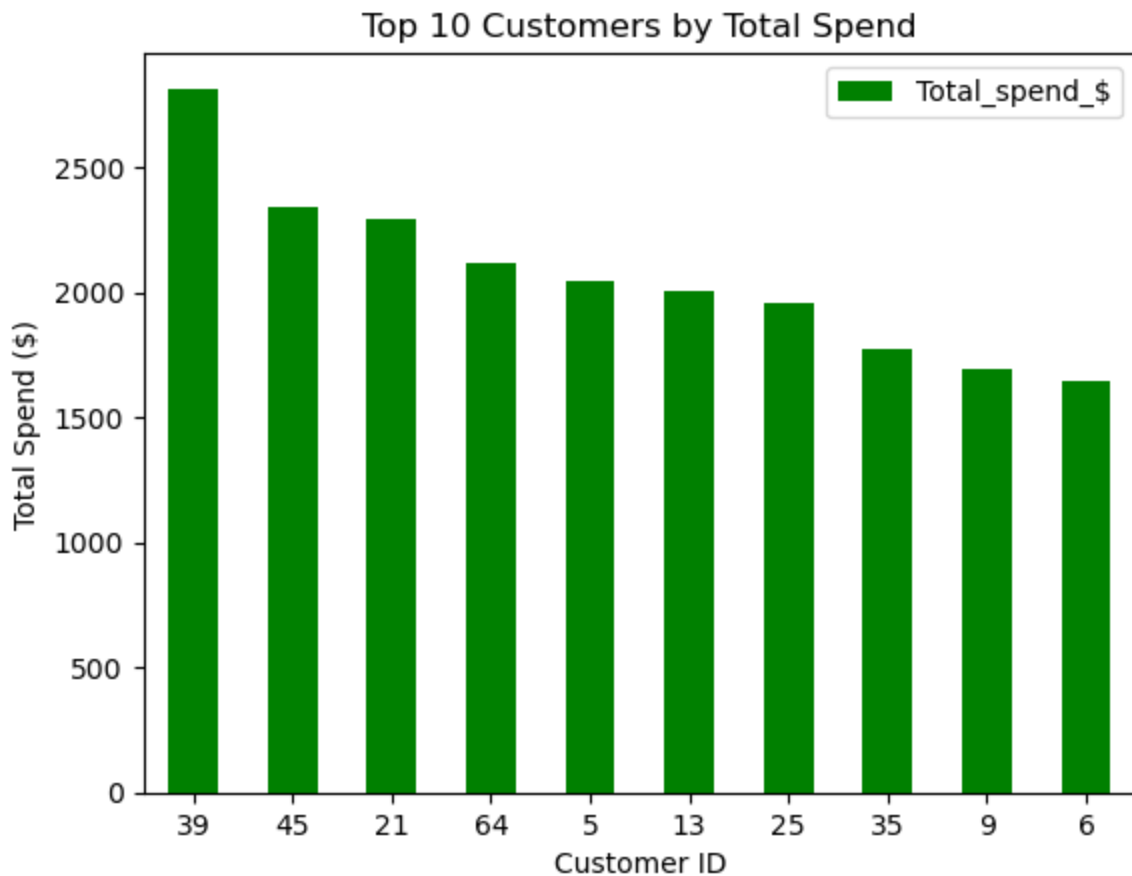
# Convert results to a DataFrame
result1_5 = pd.DataFrame(result1_5, columns=["Customer_ID", "First_Name", "Last_Name", "Total_spend_$"])

# Print the results
print(result1_5)

# Visualization: Bar Chart for Total Spend
plt.figure(figsize=(10, 6))
result1_5.plot(x="Customer_ID", y="Total_spend_$", kind="bar", color='green', rot=0)
plt.title("Top 10 Customers by Total Spend")
plt.xlabel("Customer ID")
plt.ylabel("Total Spend ($)")
plt.show()
```

	Customer_ID	First_Name	Last_Name	Total_spend_\$
0	39	Bette	Kutch	2815.0
1	45	Mercedes	Monahan	2342.0
2	21	Lois	Kuvalis	2293.0
3	64	Lillie	Hilpert	2120.0
4	5	Maudie	Wyman	2048.0
5	13	Winfield	Konopelski	2002.0
6	25	Judson	Reynolds	1960.0
7	35	Deja	Morar	1773.0
8	9	Brady	Schiller	1696.0
9	6	Lilian	Durgan	1646.0

<Figure size 1000x600 with 0 Axes>



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In [65]: # Execute the query
mycursor.execute("SELECT Customer_ID, First_Name, Last_Name, Total_rev AS 'LTR($)', RC

# Convert results to a DataFrame
result2 = pd.DataFrame(mycursor.fetchall(), columns=["Customer_ID", "First_Name", "Las

# Close cursor and connection
#mycursor.close()
#conn.close()

# Print the results
print(result2)

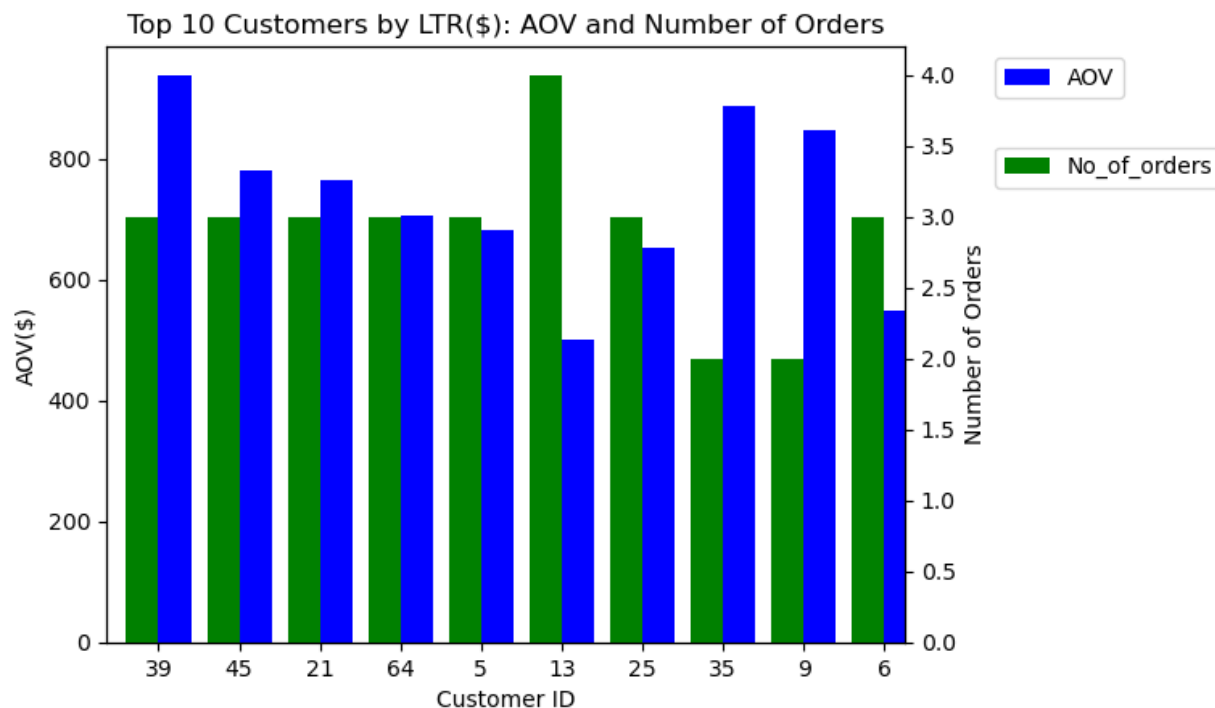
# Visualization: Bar Chart
plt.figure(figsize=(10, 6))
ax1 = result2.plot(x="Customer_ID", y="AOV($)", kind="bar", color='blue', rot=0, posit
ax2 = result2.plot(x="Customer_ID", y="No_of_orders", kind="bar", color='green', rot=0

ax1.set_title("Top 10 Customers by LTR($): AOV and Number of Orders")
ax1.set_xlabel("Customer ID")
ax1.set_ylabel("AOV($)")
ax2.set_ylabel("Number of Orders")
# Place the Legend outside the chart area
ax1.legend(["AOV"], loc="upper left", bbox_to_anchor=(1.1, 1))
ax2.legend(["No_of_orders"], loc="upper left", bbox_to_anchor=(1.1, 0.85))

plt.show()
```

	Customer_ID	First_Name	Last_Name	LTR(\$)	AOV(\$)	No_of_orders
0	39	Bette	Kutch	2815.0	938.33	3
1	45	Mercedes	Monahan	2342.0	780.67	3
2	21	Lois	Kuvalis	2293.0	764.33	3
3	64	Lillie	Hilpert	2120.0	706.67	3
4	5	Maudie	Wyman	2048.0	682.67	3
5	13	Winfield	Konopelski	2002.0	500.50	4
6	25	Judson	Reynolds	1960.0	653.33	3
7	35	Deja	Morar	1773.0	886.50	2
8	9	Brady	Schiller	1696.0	848.00	2
9	6	Lilian	Durgan	1646.0	548.67	3

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In []: